Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Project: **KebunSah**

a. Team name: **[add team name]**
b. Team leader name: **[add team leader name]**
c. Problem Statement: **Sustainability & Social Impact.** Malaysian oil-palm smallholders must show where their fruit was grown (a plot boundary and no deforestation after 2020) for the EU Deforestation Regulation, yet keep their records on paper. KebunSah turns photos of their documents and a drawn boundary into a green / yellow / red check and a plain-language to-do list.

Brief about the idea

KebunSah: a phone check that shows what is missing for EUDR

The problem

30 Dec 2026 EUDR starts for large and medium operators; micro and small operators follow on 30 Jun 2027.

450,000+ oil-palm smallholders in Malaysia, most with paper records and no digital tools.

55–65% of independent smallholder area projected to have compliance-ready data by early 2026.

Being on an official register is not proof. Registers hold names, not boundaries; a land title proves ownership, not forest history; and the documents often disagree with each other.

Sources: EU Commission, PSQR, TraceX, GLOBOIL (see notes)

The idea

Photograph the land title (several pages if needed), MPOB licence and MSPO certificate, from camera or gallery.

Draw the boundary on a satellite map, walk the corners with GPS, or mark one point for plots under 4 ha.

Get a verdict in Malay or English, with a to-do list: what is missing, why, and where to get it.

COMPLETE | NEEDS REVIEW | ACTION NEEDED

Smallholder phone app

Field officer adds land for members

Cooperative admin desktop dashboard

Opportunities

Built for the smallholder's phone, not the exporter's office

How different is it from any of the other existing ideas?

- Most EUDR tools are built for exporters, mills and large operators. KebunSah starts from the smallholder's phone, in Malay first.
- It checks the documents against each other and against the drawn boundary: a title proves ownership, not forest history, and registers hold names, not boundaries.
- It tells the farmer what to do next (what is missing, why, where to get it) instead of only reporting a score.

How will it be able to solve the problem?

- Photos of the land title, MPOB licence and MSPO certificate become structured data (Gemini), each field with a confidence score.
- Rules compare area, expiry dates and names; Earth Engine screens the plot for forest loss after 31 Dec 2020.
- The result is a green / yellow / red verdict and a to-do list. Cooperatives follow every member plot and export dossiers for buyers.

USP of the proposed solution

- **Gemini reads and explains; rules decide.** Every verdict is reproducible and auditable.
- **Made for the field:** large buttons, status as colour + icon + words, type or speak a question, and it keeps working with no signal.
- **Free for smallholders;** cooperatives, mills and buyers pay, because they carry the due-diligence duty.

List of features offered by the solution

Nine features in one chain: capture, verdict, dossier

Photograph documents
Land title (up to 4 pages), MPOB licence and MSPO certificate, from camera or gallery.

Draw the boundary
On a satellite map, by walking the corners with GPS, or one point plus area under 4 ha.

AI document reading
Gemini reads each photo into fields with a confidence score; unsure ones go back to the user.

Rules check
Area (title vs drawing), expiry dates and names are compared across the documents.

Forest screening
Earth Engine looks for forest loss after 31 Dec 2020 inside the boundary. Marked preliminary.

Verdict and to-do list
Green, yellow or red, plus what is missing, why, and where to get it. A step log shows how.

Ask by typing or voice
In Malay or English; answers only about the plot, the EUDR and the documents needed.

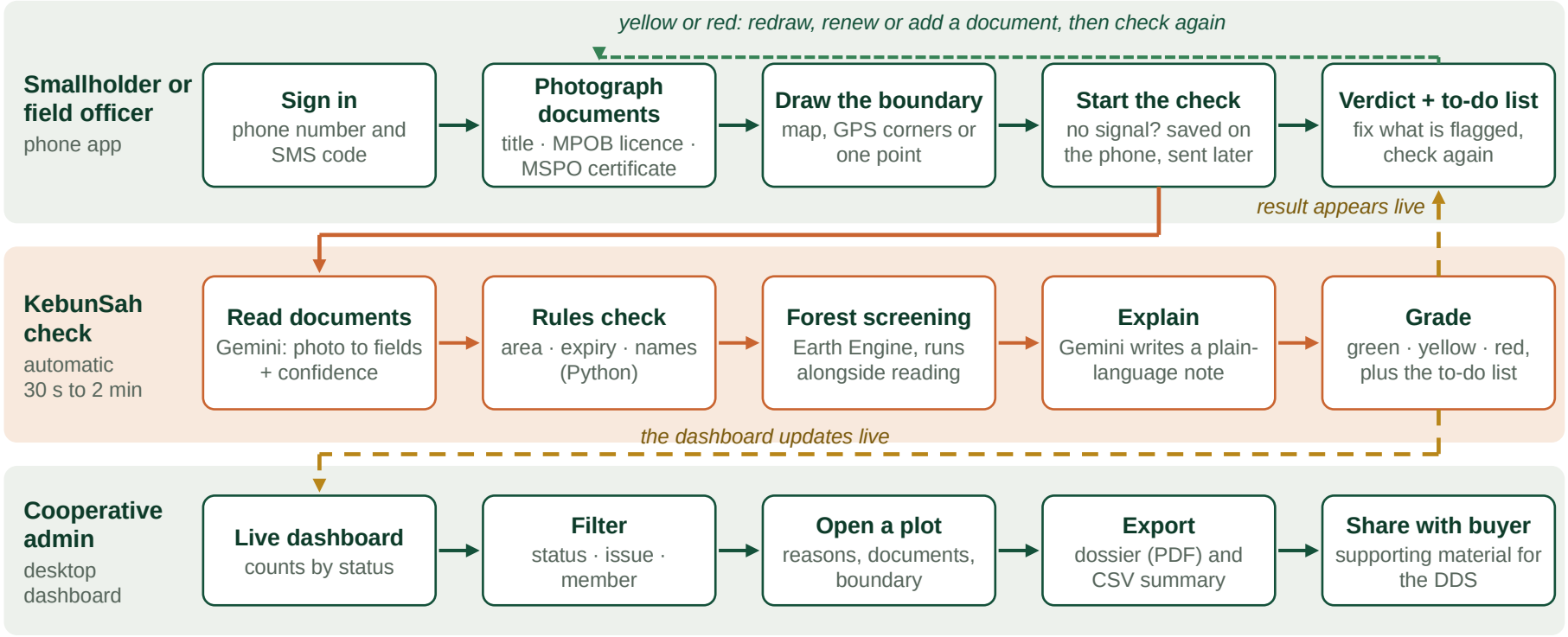
Works with no signal
Drafts are kept on the phone and sent automatically when the signal returns.

Cooperative dashboard
Every member plot by status and issue, a CSV summary and a printable dossier per plot.

Three roles: smallholder · field officer (adds land on a member's behalf) · cooperative admin (read-only). Languages: Malay and English.

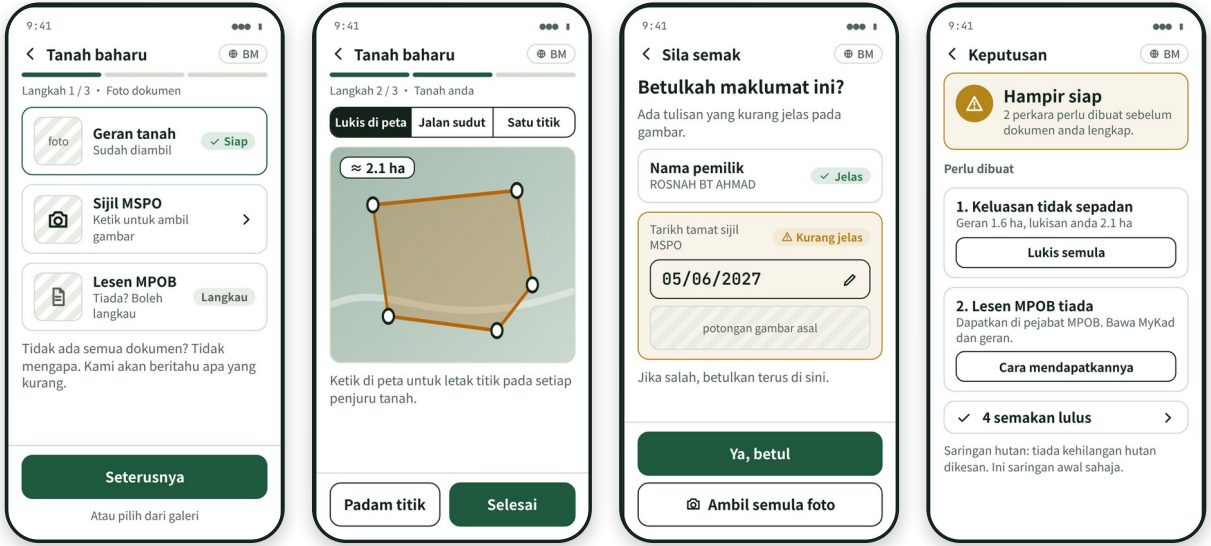
Process flow diagram or Use-case diagram

From a photo to a verdict in about two minutes



Wireframes/Mock diagrams of the proposed solution (optional)

Designed for low digital skill: one screen, one task



1 Photograph the documents | **2 Draw the boundary** | **3 Confirm what AI could not read** | **4 Verdict and to-do list**

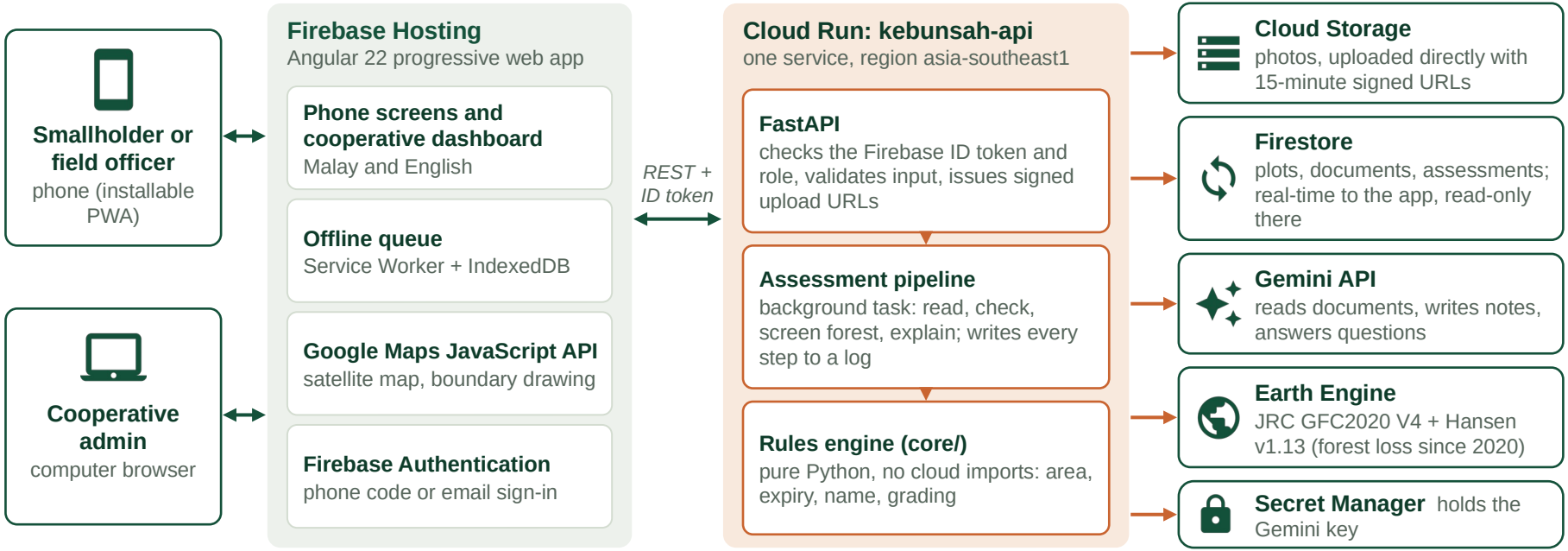
Low-fidelity wireframes from the design phase (screen text in Malay). The built screens are on the snapshot slides.

Design rules

- **One screen, one task.** One main button and a progress bar (Step 1 / 3).
- **Big buttons in the thumb zone.** Touch targets of at least 48 px.
- **Malay first.** English one tap away; icons always with words.
- **Status is never colour alone.** Colour, icon and words together.
- **Plain, kind words.** "Take the picture again", not "Recognition failed".
- **Forgiving.** Skip a document, retake a photo, fix what AI read, leave while it checks.

Architecture diagram of the proposed solution

One Cloud Run service, with Gemini and Earth Engine as tools



CI/CD GitHub → Cloud Build → Artifact Registry → Cloud Run and Firebase Hosting. Cloud Run keeps no idle instance (min 0, max 3).

Next Cloud Tasks + separate AI worker, Vertex AI quotas, cooperative-scoped Firestore rules.

Technologies to be used in the solution

Google Cloud AI at the core, standard open tools around it

Google AI

- Gemini 3.5 Flash, with fallback to other Flash and Flash-Lite models
- Photo reading: JSON schema output, temperature 0, confidence per field
- Notes and typed or spoken Q&A
- Earth Engine forest-loss datasets

Google Cloud and Firebase

- Cloud Run (scales to zero), Cloud Build, Artifact Registry
- Firebase Hosting and Authentication (phone code, email)
- Firestore, Cloud Storage (signed URLs), Secret Manager
- Maps JavaScript API for the satellite map

Front end

- Angular 22, TypeScript, installable PWA
- Service Worker and IndexedDB offline queue
- Turf.js area preview; MediaRecorder voice input; browser speech output
- Vitest: 993 tests

Back end

- Python 3.11, FastAPI, Pydantic (OpenAPI docs built in)
- google-genai, earthengine-api, firebase-admin
- Shapely + pyproj: area on the ellipsoid, server-side
- RapidFuzz name similarity; pytest: 847 tests

Reliability and safety

- Rules in pure Python: same input, same answer
- Retry, back-off and model fallback on API errors
- A failed step becomes "needs review"
- Read-only browser; IDs stored only as a keyed hash

Delivery

- GitHub + Cloud Build: a push to main deploys API and app
- 200 synthetic demo plots, watermarked sample documents
- Full journey driven in a real browser on the deployed service
- Public repo, English README, judge access

Estimated implementation cost (optional)

Cents per check, and free tiers carry the prototype

Cost item	Prototype today	Pilot: one cooperative, ~1,000 plots
Gemini API	Free tier or pay-as-you-go; est. under US$10 for build, tests and judging	≈ US$30–60 per full pass of checks (≈ 3–6 US cents per check: 3 photo reads + 1 note)
Cloud Run	Within free-tier limits at demo traffic; no idle instance (min 0, max 3)	Free tier to low tens of US$ per month
Firestore, Storage, Hosting, Auth	Free tier or cents	Low US$ per month; ~1.2 GB of photos per 1,000 plots
Google Maps JavaScript API	Free monthly allowance	Beyond the allowance ≈ US$7 per 1,000 map loads
Earth Engine	Non-commercial use	A commercial deployment needs a paid plan (from US$500 per month)
People	2 builders, 10 days	Cooperative onboarding and support

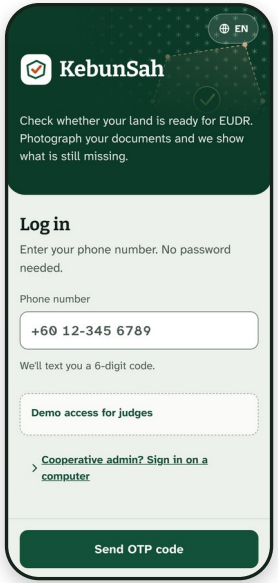
Estimates from public list prices (Gemini 3.5 Flash US$1.50 / US$9.00 per 1M input / output tokens, third-party trackers, 18 Oct 2026) and token estimates. Not a quote; verify on Google's pricing pages.

Cost guardrails built in

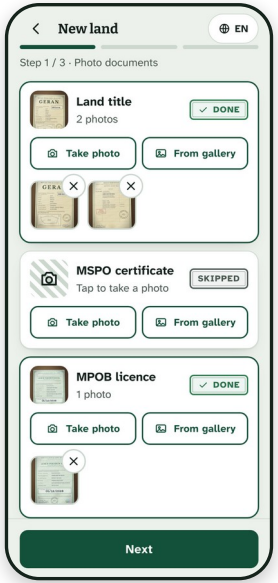
- **Flash-class models only.** Reading a photo does not need a Pro model.
- **Up to four Gemini calls per check,** and documents already read are reused on a re-check.
- **Photos shrunk on the phone** to 1,600 px JPEG before upload.
- **A hard ceiling:** Cloud Run runs at most 3 instances and none when idle.
- **Fallback models and retries** absorb quota errors instead of repeating paid calls.
- **Token usage is logged** for each Gemini call, so spend can be watched.

Snapshots of the prototype

The phone app: photograph, draw, check, fix



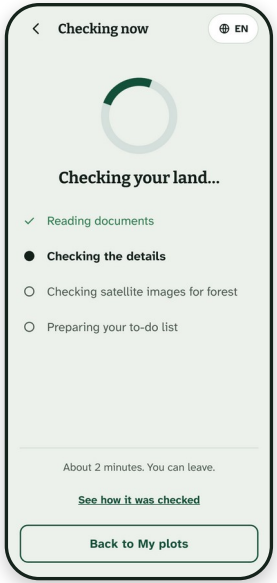
1 Sign in
phone number and code; Malay or English



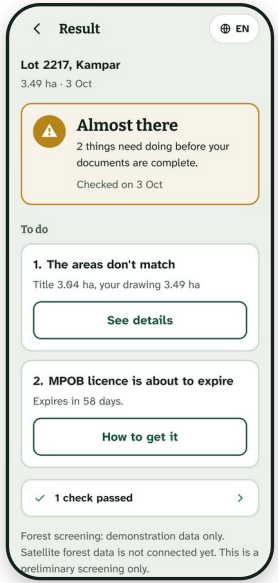
2 Photograph
land title (2 pages), MPOB licence



3 Draw
boundary on a satellite map, about 3.29 ha



4 Check
steps tick off as they finish



5 Verdict
yellow: two things to do

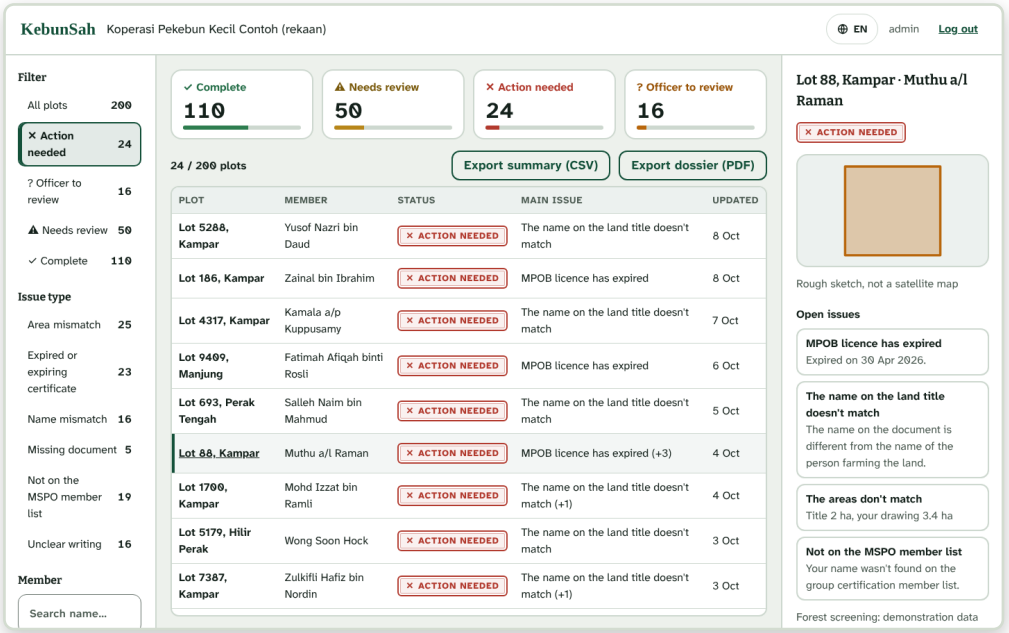


6 Why
title 3.04 ha, drawing 3.49 ha

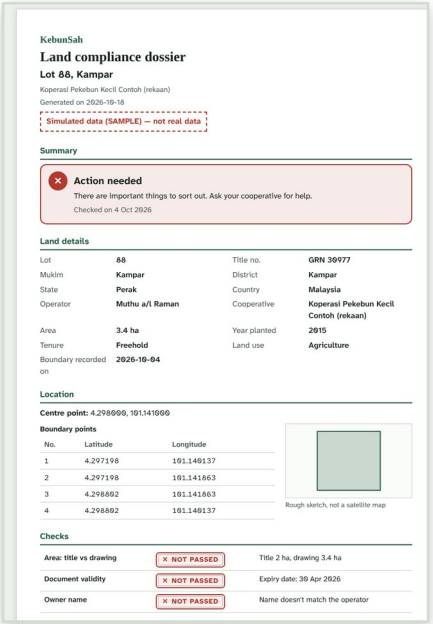
Working prototype with synthetic demo data (documents are watermarked SIMULATED). Seeded plots carry pre-computed forest values; new checks call Earth Engine.

Snapshots of the prototype

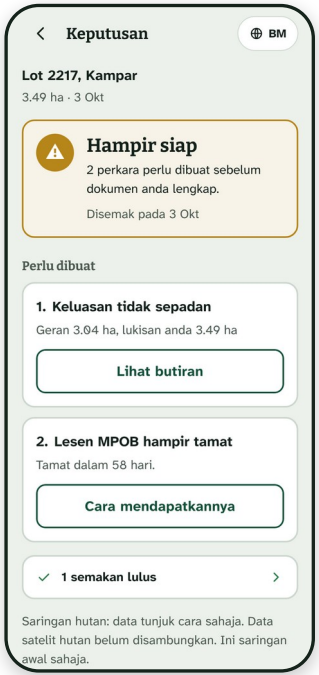
The cooperative dashboard, the dossier and Malay-first screens



Cooperative dashboard (computer)
200 member plots by status and issue, with reasons and exports.



Dossier per plot
A4 layout, saved as PDF



Malay first
one tap to English

Working prototype with synthetic demo data. The cooperative admin role is read-only.

Prototype Performance report/Benchmarking

Measured on the submitted build: 1,840 tests pass, 40 of 40 fields read

1,840
automated tests pass: 847 back end (pytest) and 993 front end (Vitest)

40 / 40
fields read correctly by Gemini on 16 synthetic photos, with handwriting, blur, tilt and low light

200 / 200
demo plots get the same verdict from the Python rules engine as from the independent seed generator

30 s – 2 min
from tap to verdict with real Gemini calls (about 30 s locally; up to 2 min on free-tier quota)

What we measured	How	Result
Rules engine	434 tests: area 10% or 0.1 ha, expiry 90 days, name match 0.85, confidence 0.7	pass
AI pipeline	295 tests: quota retry, back-off, model fallback, a failed step degrades to review	pass
API	118 tests: sign-in, roles (cooperative admin read-only), signed URLs, assess, delete, voice	pass
Front end	993 tests; browser code has no Firestore writes; production bundle has no emulator traces	pass
Phone app weight	Production build, initial load: 485 kB raw, about 129 kB compressed	measured

Run on 18 Oct 2026, commit c9e762c. Gemini accuracy: team smoke test (scripts/gemini_smoke.py).

Known limits

- Free-tier Gemini allows 5 requests a minute per model, so back-to-back checks slow down.
- Forest data are public global datasets: a preliminary screening; small plots can be misjudged.
- Seeded demo plots carry pre-computed forest values, labelled as such.
- The satellite map needs signal; walking the corners and one-point entry work offline.

Additional Details/Future Development (if any)

Free for smallholders; those who carry the EUDR duty pay

Business model

Who pays	How	What they get
Cooperatives	Annual fee per member plot	All members' compliance in one place; dossiers for buyers
Mills and dealers	Subscription by fruit volume or supplier plots	Traceable fruit, fewer rejections by EU buyers
Exporters and EU buyers	API access fee	Upstream compliance dossiers delivered directly
Government and NGO programmes	Project partnerships	Smallholder digitisation at scale

Why the buyers pay: under the EUDR the due-diligence duty sits with operators and traders, who need clean plot data from thousands of smallholders. KebunSah is how they get it without visiting every farm.

Impact

- 450,000+ Malaysian oil-palm smallholders can learn what is missing before a buyer turns their fruit away.
- Cooperatives see risk across all members in one view, and can act early.

Responsible AI

- A preliminary screening, not legal advice or certification; results say so.
- Gemini reads and explains; rules decide. Unsure fields go back to a person.
- Synthetic demo data only; ID numbers are never stored in full.

Additional Details/Future Development (if any)

From Malaysian palm oil to every EUDR commodity

Today

Malaysia palm oil

Prototype live; three document types; Malay and English; synthetic data.

Next

Pilot with a cooperative

Masked real documents from 2–3 smallholders; verify fields against e-MSPO and E-LesenPK.

Then

Indonesia and Thailand

Palm smallholders; language and land-office process become configuration.

Later

Other EUDR commodities

Rubber, cocoa, coffee, timber: new document types and rules, same app.

Engineering next

- Cloud Tasks and a separate AI worker for volume
- Vertex AI for production quotas and data residency
- Firestore rules scoped to each cooperative and farmer
- Regulation text retrieval, with citations in the notes
- Official-site verification (e-MSPO, MPOB E-LesenPK)
- Mill ERP integration and chain of custody
- Sabah and Sarawak land systems
- Admin override with a reason; server-side PDF

Why it scales

- Thresholds already live in a config document, not in code.
- A new commodity means new document types and rules, not a new app.
- The rules engine has no cloud imports, so it can move to any service.

Provide links to your:

Try it, read it, watch it

GitHub Public Repository	**github.com/longrong1110/KebunSah**	Source code, English README, tests, API contract and the 200-plot demo data generator.
Demo Video Link (3 Minutes)	**[paste the YouTube / Vimeo / Drive link here]**	Narrated in English: problem, farmer's phone, check, cooperative dashboard, impact.
Final Product Link	**kebunsah.web.app**	Open on a phone. On the sign-in page use "Demo access for judges".

Judge access (all data is synthetic)

Phone app

Open the Final Product Link on a phone, tap "Demo access for judges". No SMS is sent; the code is 123456.

+60 10-000 0011 smallholder, mostly green
+60 10-000 0012 yellow plot
+60 10-000 0013 red plot
+60 10-000 0001 field officer

Cooperative dashboard (computer)

Open **kebunsah.web.app/coop/login** and use the demo box on that page.

Email **koperasi@kebunsah.demo**
Password **Koperasi-Demo-2026**

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Build in JAPAC.
Build for the world.

Thank you

